

Data Profiling Report

Project Title: Customer Analytics Platform for Strategic Customer Retention and Revenue Growth

Dataset: Olist Brazilian E-commerce Dataset

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1. Executive Summary

Purpose

The objective of this data profiling exercise is to develop an initial understanding of the structure, scale, completeness, uniqueness, distributions, and business characteristics of the Olist e-commerce dataset before performing data quality assessment and transformation into the Silver layer.

Key findings

- The dataset contains approximately **1.55 million records across 9 tables**.
- The order dataset covers orders for the time period of 2 years and 1 month from **04 September 2016 to 17 October 2018**.
- The customer dataset contains **99,441 unique customer IDs and 96,096 unique customer identities**.
- **97%** of orders are in the delivered status.
- Payment data indicates **credit cards** are the most common payment method.
- Several tables exhibit expected one-to-many relationships, particularly **orders -> order items** and **orders -> payments**.

2. Dataset Overview

2.1 Table Inventory

Table	Business Entity	Rows	Key Identifier	Grain
Customers	Customer	99,441	customer_id	One row per customer/order relationship
Orders	Order	99,441	order_id	One row per order
Order Items	Order Line Item	1,12,650	order_id + order_item_id	One row per item within an order
Order Payments	Payment	1,03,886	order_id + payment_sequential	One row per payment transaction
Order Reviews	Review	99,224	review_id	One row per review
Products	Product	32,951	product_id	One row per product
Sellers	Seller	3,095	seller_id	One row per seller
Geolocation	Geographic Record	10,00,163	ZIP prefix + location	Multiple records per ZIP prefix

Table	Business Entity	Rows	Key Identifier	Grain
Category Translation	Category	71	product_category_name	One row per category

3. Data Coverage

3.1 Temporal Coverage

Metric	Result
Earliest order	2016-09-04
Latest order	2018-10-17
Observation period	2 years 1 month

Observation

The Olist transactional dataset covers the period from September 4, 2016 to October 17, 2018, representing approximately 2 years and 1 month of observed order activity across three calendar years. The customer master table does not contain customer registration or acquisition dates; therefore, customer-level temporal analysis will be based on transaction activity rather than customer creation dates.

4. Customer Data Profile

4.1 Customer Characteristics

Metric	Result
Total customer records	99,441
Distinct customer_id	99,441
Distinct customer_unique_id	96,096
Distinct cities	4,119
Distinct states	27

4.2 Geographic Distribution (Top 10 States)

State	Customers	% of Customers
SP	41,746	41.98
RJ	12,852	12.92
MG	11,635	11.70
RS	5,466	5.50
PR	5,045	5.07
SC	3,637	3.66
BA	3,380	3.40
DF	2,140	2.15
ES	2,033	2.04

State	Customers	% of Customers
GO	2,020	2.03

Key observations

- **Customer ID uniqueness:** The dataset contains 99,441 customer records and 99,441 distinct customer_id values, indicating that customer_id is unique within the customer table.
- **Multiple customer identifiers:** There are 96,096 distinct customer_unique_id values compared with 99,441 customer_id values. This indicates that some customer identities are associated with multiple customer_id records and warrants further investigation before defining the customer grain for downstream analytics.
- **Geographic coverage:** Customers are distributed across 27 Brazilian states and 4,119 cities, providing broad geographic coverage for regional analysis.
- **High geographic concentration:** São Paulo accounts for 41.98% of all customer records, while Rio de Janeiro and Minas Gerais account for 12.92% and 11.70%, respectively. The three states together represent approximately 66.6% of the customer base.

Business relevance

The distinction between customer_id and customer_unique_id is important for customer-level analytics. Metrics such as purchase frequency, retention, RFM segmentation, churn, and customer lifetime value must be calculated at an appropriately defined customer grain to avoid misrepresenting customer behavior.

The strong geographic concentration of customers, particularly in São Paulo, Rio de Janeiro, and Minas Gerais, should be considered when evaluating regional customer behavior, sales performance, delivery experience, and retention. Overall business-level metrics may be disproportionately influenced by these major markets, making regional comparisons important for identifying geographic differences in customer behavior and performance.

5. Order Data Profile

5.1 Order Volume

Metric	Result
Total orders	99,441
Distinct orders	99,441
Distinct customers	99,441
First order	2016-09-04 21:15:19
Last order	2018-10-17 17:30:18

5.2 Order Status Distribution

Order Status	Orders	% of Orders
delivered	96,478	97.02
shipped	1,107	1.11
cancelled	625	0.63
unavailable	609	0.61

Order Status	Orders	% of Orders
invoiced	314	0.32
processing	301	0.30
created	5	0.01
approved	2	0.00

Key observations

- **Order volume:** The dataset contains 99,441 orders, with the same number of distinct order_id values, indicating that order_id is unique in the orders table.
- **Order-customer relationship:** There are 99,441 distinct customer_id values for 99,441 orders, indicating that each customer_id appears once in the order table. This should be interpreted alongside the distinction between customer_id and customer_unique_id identified in the customer table.
- **Temporal coverage:** Orders span from September 4, 2016 to October 17, 2018, providing approximately 2 years and 1 month of transaction history.
- **Very high delivery completion:** 97.02% of orders are marked as delivered, while only 2.98% fall into other statuses.
- **Small proportion of unsuccessful/incomplete orders:** Canceled orders account for 0.63%, while unavailable orders account for 0.61%. Shipped orders represent another 1.11%, indicating that the overwhelming majority of transactions reached the delivered stage.
- **Long-tail order statuses:** Invoiced, processing, created, and approved orders together account for less than 1% of orders. These low-frequency states will need to be considered carefully when constructing operational or delivery-performance metrics.

Business relevance

The high proportion of delivered orders makes the dataset particularly suitable for analyzing completed customer journeys, including purchasing behavior, delivery experience, customer satisfaction, and repeat purchasing.

The presence of multiple order statuses allows the project to examine the order lifecycle and operational performance, rather than treating every transaction as a completed sale. Status-specific handling will be important when calculating metrics such as delivery time, cancellation rate, fulfillment performance, and customer retention.

The approximately two-year transaction history provides sufficient temporal variation for trend analysis and first-purchase cohort analysis, although the absence of customer registration dates means that customer tenure must be derived from transaction activity rather than account creation.

6. Transaction & Revenue Profile

6.1 Order Items

Metric	Result
Total order items	112,650
Distinct orders	98,666
Distinct products	32,951
Distinct sellers	3,095

Metric	Result
Minimum item price	0.85
Median item price	74.99
Mean item price	120.65
Maximum item price	6,735.00

6.2 Freight

Metric	Result
Minimum freight	0.00
Median freight	16.26
Mean freight	19.99
Maximum freight	409.68

Key observations

- **Order-item volume exceeds order volume:** The dataset contains 112,650 order items across 98,666 distinct orders, indicating that many orders contain multiple products. Therefore, the order-items table operates at line-item level rather than order level.
- **Broad product and seller coverage:** The order items involve 32,951 distinct products and 3,095 sellers, providing substantial variation for product and seller performance analysis.
- **Highly right-skewed item prices:** The mean item price (120.65) is substantially higher than the median (74.99), suggesting that the distribution is positively skewed and that relatively high-priced products influence the average.
- **Large price range:** Item prices range from 0.85 to 6,735.00, indicating substantial variation in product price points.
- **Freight costs are also right-skewed:** Mean freight cost (19.99) is higher than the median (16.26), suggesting that a smaller number of high-freight shipments increase the average.
- **Wide variation in freight costs:** Freight ranges from 0 to 409.68, indicating substantial differences in shipping costs across orders/products.

Business Relevance

The difference between order-level and item-level grain is important when calculating revenue, average order value, product performance, and customer spending. Order-item records must be appropriately aggregated to the order or customer level to avoid double-counting.

The right-skewed distribution of both product prices and freight costs suggests that median and percentile-based measures should complement averages when evaluating customer spending, product pricing, and logistics costs.

The broad range of products and sellers provides an opportunity to analyze product/category performance and seller contribution, while the variation in freight costs can be used to investigate the relationship between shipping costs, product characteristics, geographic location, customer experience, and potentially repeat purchasing.

7. Payment Profile

7.1 Payment Methods

Payment Type	Transactions	%
credit_card	76,795	73.92
boleto	19,784	19.04
voucher	5,775	5.56
debit_card	1,529	1.47
not_defined	3	0.00

7.2 Payment Amount

Metric	Result
Minimum	0.00
Median	100.00
Mean	154.10
Maximum	13,664.08

Key observations

- **Credit cards dominate payment activity:** Credit card payments account for 73.92% of payment transactions, followed by boleto at 19.04%. Voucher and debit card payments represent relatively small shares.
- **Payment values are highly right-skewed:** The mean payment value (154.10) is substantially higher than the median (100.00), indicating that a relatively small number of high-value transactions are pulling the average upward.
- **Wide payment-value range:** Payment values range from 0.00 to 13,664.08, indicating substantial variation in transaction values and the presence of potentially unusual zero-value payments that should be investigated during data quality assessment.
- **Payment table is transaction-level rather than necessarily order-level:** The number and structure of payment records should be examined for orders with multiple payment records before calculating order-level revenue or payment metrics.

Business Relevance

The strong dominance of credit-card payments indicates that credit cards are the primary payment channel in the observed customer base. Payment method can therefore be incorporated into customer segmentation and behavioral analysis to identify differences in purchasing patterns across payment types.

The substantial difference between median and mean payment values suggests that median and percentile-based measures should complement averages when evaluating customer spending and transaction value.

The presence of zero-value payments and very high payment values warrants further data-quality investigation. These records should be validated against the underlying order and payment structure before being included in revenue or customer-value calculations.

The payment data can also support analysis of payment behavior, transaction value, and customer value, but payment records must be appropriately aggregated at the order/customer level to avoid double-counting.

8. Product Profile

8.1 Product Characteristics

Metric	Result
Total products	32,951
Distinct categories	73
Minimum weight	0
Median weight	700.00
Mean weight	2,276.47
Maximum weight	40,425

8.2 Top Product Categories

Rank	Category	Products
1	cama_mesa_banho	3,029
2	esporte_lazer	2,867
3	moveis_decoracao	2,657
4	beleza_saude	2,444
5	utilidades_domesticas	2,335

Observations

- **Broad product assortment:** The dataset contains 32,951 products across 73 categories, providing substantial product-level variation for assortment and category analysis.
- **Product weight is highly right-skewed:** The mean weight (2,276.47 g) is substantially higher than the median (700 g), indicating that a relatively small number of heavy products significantly influence the average.
- **Wide variation in product weight:** Product weights range from 0 g to 40,425 g (40.4 kg). The zero-weight records should be investigated during the data-quality assessment.
- **Category concentration:** The largest product categories are cama_mesa_banho (bed, bath & table), esporte_lazer (sports & leisure), moveis_decoracao (furniture & decoration), beleza_saude (health & beauty), and utilidades_domesticas (household utilities). These categories collectively represent a substantial portion of the product catalogue.

Business Relevance

The breadth of the product catalogue enables analysis of category-level customer preferences, product performance, assortment concentration, and cross-category purchasing behavior.

The strong right-skew in product weight is relevant to logistics analysis because heavier products may generate disproportionately higher freight costs. Product weight can therefore be incorporated into analysis of shipping costs, delivery performance, and customer experience.

The presence of products with zero recorded weight requires validation before using weight as a logistics or product-level analytical variable. If these values represent missing or incorrectly recorded measurements rather than genuinely weightless products, they should be handled appropriately in the Silver layer.

The Portuguese category names can be mapped to their English equivalents using the category translation table, improving interpretability for downstream analysis and reporting.

9. Customer Experience Profile

9.1 Review Scores

Score	Reviews	%
1	11,424	11.51
2	3,151	3.18
3	8,179	8.24
4	19,142	19.29
5	57,328	57.78

9.2 Review Text Availability

Metric	Count	%
Total reviews	99,224	100%
Reviews with title	11,568	11.00
Reviews with message	40,977	41.00
Reviews without message	58,247	58.00

Observations

- **Strongly positive review distribution:** 57.78% of reviews have a score of 5, while another 19.29% have a score of 4. Thus, 77.07% of reviews are positive (4–5).
- **Meaningful low-score segment:** 11.51% of reviews have a score of 1, while scores of 1–2 together account for 14.69% of reviews. This provides a sufficiently large dissatisfied-customer segment for analyzing negative customer experiences.
- **Limited middle-ground ratings:** Only 8.24% of reviews have a score of 3, suggesting that customers in this dataset tend to provide relatively polarized ratings rather than concentrating around the midpoint.
- **Review text is incomplete:** Only 41% of reviews contain a review message, while approximately 59% do not. Review titles are even less common, with only 11% containing a title.

Business Relevance

The large proportion of 4- and 5-star reviews indicates generally positive reported customer experiences. However, the 14.69% share of 1- and 2-star reviews represents a substantial dissatisfied customer segment that can be analyzed to identify potential drivers of poor customer experience.

Review scores can be incorporated into the Customer Intelligence Platform as a measure of customer satisfaction, alongside behavioral variables such as purchase frequency, spending, and recency. This allows us to examine whether customer satisfaction is associated with repeat purchasing and customer value.

The substantial amount of missing review text limits the use of text-based sentiment or topic analysis to a subset of customers. Therefore, review scores provide more complete coverage for quantitative customer-experience analysis, while review text should be treated as an optional supplementary source rather than a universal customer attribute.

The high proportion of 5-star ratings should also be considered when interpreting customer satisfaction metrics, as the distribution may indicate rating skew rather than uniformly strong customer experiences.

10. Seller Profile

Metric	Result
Total sellers	3,095
Distinct seller states	23
Largest seller state	SP
Sellers in largest state	1,849

Observations

- **Seller base:** The dataset contains 3,095 sellers, providing a substantial seller population for marketplace-side analysis.
- **Geographic concentration:** São Paulo is the dominant seller location, with 1,849 sellers, representing approximately 59.74% of all sellers.
- **Seller concentration in São Paulo:** The concentration of sellers in São Paulo is even stronger than the concentration of customers there (41.98%), indicating that São Paulo is an important marketplace hub on both the demand and supply sides.

Business Relevance

The strong concentration of sellers in São Paulo is relevant when evaluating seller performance, geographic supply availability, delivery costs, and fulfillment efficiency. Regional differences in seller concentration may help explain variations in delivery experience and product availability.

Seller-level analysis can subsequently be used to identify high-performing sellers, seller concentration by category, revenue contribution, and relationships between seller characteristics and customer satisfaction.

11. Geolocation Profile

Metric	Result
Total records	1,000,163
Distinct ZIP prefixes	19,015
Distinct cities	8,011
Distinct states	27

Key observation

- **Large geographic dataset:** The geolocation table contains 1,000,163 records, making it substantially larger than the other tables in the dataset.
- **Multiple records per ZIP prefix:** There are only 19,015 distinct ZIP-code prefixes compared with more than one million records. Therefore, the geolocation table is not at ZIP-prefix grain and contains multiple geographic observations for individual ZIP prefixes.
- **Broad geographic coverage:** The dataset covers 8,011 cities and all 27 Brazilian states, providing extensive geographic coverage.

- **Potential aggregation requirement:** Because multiple records can correspond to the same ZIP prefix, the table cannot be directly joined to customer or seller data without first determining an appropriate geographic aggregation strategy.

Business Relevance

The geolocation data provides an opportunity to incorporate geographic information into customer, seller, and logistics analysis. However, its finer geographic grain requires careful treatment before integration with the analytical model.

An inappropriate join could multiply customer, order, or revenue records, leading to materially incorrect business metrics. The geolocation table will therefore require explicit transformation and validation in the Silver layer before being used for downstream analysis.

12. Missing-Value Profile

Table	Column	Total Rows	Missing	Missing %
Orders	Order approved at	99,441	160	0.16
Orders	Carrier date	99,441	1,783	1.79
Orders	Customer delivery date	99,441	2,965	2.98
Reviews	Review comment title	99,224	87,656	88.34
Reviews	Review comment message	99,224	58,247	58.7
Products	Category	32,951	610	1.85
Products	Weight	32,951	2	0
Products	Dimensions	32,951	2	0

Key Observations

- **Order approval timestamps:** `order_approved_at` has only 160 missing values (0.16%), indicating very high completeness. These records should nevertheless be investigated against order status to determine whether missing approval timestamps are consistent with orders that were canceled or otherwise not processed.
- **Carrier delivery timestamps:** `order_delivered_carrier_date` is missing for 1,783 orders (1.79%). The missingness may be associated with orders that had not progressed to the carrier-delivery stage and should therefore be assessed against order status before any imputation is considered.
- **Customer delivery timestamps:** `order_delivered_customer_date` is missing for 2,965 orders (2.98%). Given that this is approximately the same proportion as the non-delivered orders identified in the order-status analysis, the missingness may largely reflect orders that were not completed. This should be validated as part of the data-quality assessment.
- **Review titles:** `review_comment_title` is missing for 88.34% of reviews. Since providing a review title is optional, this is more appropriately interpreted as limited customer-provided text rather than automatically treated as a data-quality problem.
- **Review messages:** `review_comment_message` is missing for 58.70% of reviews. As review text is optional, these missing values are likely to represent customers who submitted a rating without providing written feedback.
- **Product categories:** 610 products (1.85%) have a missing product category. Since product category is important for product and customer-preference analysis, these records require investigation and potential mapping using the category translation/reference data.

- **Product physical attributes:** Only 2 products have missing weight and 2 products have missing dimensions. The extremely low level of missingness suggests that these fields are largely complete, although the affected records should be handled appropriately during transformation.

Business Relevance

Missingness in operational fields such as delivery timestamps can directly affect metrics such as delivery time, fulfillment performance, and customer experience. These fields should therefore be handled according to order status rather than through indiscriminate imputation.

The high level of missingness in review text limits the coverage of text-based customer-experience analysis. Review scores, which are substantially more complete, will therefore be a more reliable quantitative measure of customer satisfaction.

Missing product categories may affect category-level sales and customer-preference analysis. These records should be investigated before the category variable is used for segmentation or performance analysis.

13. Initial Data Observations

Observation 1 — Customer identity and analytical grain

customer_id and customer_unique_id exhibit different levels of identification, with 99,441 distinct customer_id values compared with 96,096 distinct customer_unique_id values. This distinction must be resolved when defining the customer grain for downstream metrics such as purchase frequency, retention, RFM segmentation, churn, and customer lifetime value.

Observation 2 — Multiple levels of transaction grain

The dataset contains entities at different granularities. Orders are recorded at order level, while order items and payments can contain multiple records associated with a single order. Consequently, order-level and customer-level metrics cannot be calculated by directly aggregating all tables together without first establishing appropriate aggregation rules.

Observation 3 — Payment aggregation

The payment data contains multiple payment records for some orders, indicating that payment information operates at a finer grain than the order table. Payment values will therefore need to be aggregated appropriately before being incorporated into order- or customer-level financial metrics.

Observation 4 — Geographic structure

The geolocation table contains more than one million records but only 19,015 distinct ZIP-code prefixes, indicating that multiple geographic observations exist for individual ZIP prefixes. The table therefore requires an explicit aggregation strategy before being joined with customer or seller data to prevent unintended record multiplication.

Observation 5 — Geographic concentration

Customer and seller activity is geographically concentrated, with São Paulo representing approximately 42% of customers and nearly 60% of sellers. This concentration should be considered when interpreting overall marketplace performance and when investigating regional differences in customer behavior, seller availability, delivery performance, and logistics costs.

Observation 6 — Customer experience

Customer reviews are strongly concentrated toward positive ratings, with 77.07% of reviews receiving scores of 4 or 5. However, 14.69% receive scores of 1 or 2, providing a sufficiently large dissatisfied-customer segment for investigating drivers of poor customer experience.

Observation 7 — Missing review text

Review scores are considerably more complete than written feedback. Since approximately 59% of reviews lack a review message, quantitative satisfaction analysis should primarily rely on review scores, while text-based analysis should be treated as a supplementary analysis on the available subset.

Observation 8 — Transaction-value distribution

Both item prices and payment values exhibit right-skewed distributions, with means substantially above their respective medians. This indicates the presence of relatively high-value transactions and suggests that median and percentile-based measures should complement averages when analyzing customer spending and transaction value.

Observation 9 — Order completion

Approximately 97% of orders are marked as delivered. This provides a strong base for analyzing completed customer journeys, while the smaller population of canceled, unavailable, and in-progress orders can be used to investigate fulfillment and operational exceptions.

Observation 10 — Product and logistics characteristics

Product weight exhibits substantial dispersion, with the mean considerably above the median and a small number of missing or zero-weight observations. Product physical characteristics may therefore be relevant for analyzing freight costs and delivery performance, subject to validation of anomalous values during the data-quality stage.

14. Questions for Data Quality Assessment

The profiling exercise identified several areas requiring further validation before the data can be transformed into the Silver layer. The following checks will form the basis of the Data Quality Assessment.

ID	Data Quality Question	Why It Matters
DQ-001	Are customer_id values unique in the customer table?	Establishes the correct customer-table grain
DQ-002	Are order_id values unique in the orders table?	Establishes the correct order grain
DQ-003	Are customer_unique_id and customer_id relationships consistent?	Determines the appropriate customer identifier for customer-level analytics
DQ-004	Do all orders map to a valid customer record?	Referential integrity
DQ-005	Do all order items map to valid orders, products, and sellers?	Referential integrity across transaction tables
DQ-006	Do payment records map to valid orders?	Referential integrity and financial accuracy
DQ-007	Are payment values valid, including zero or negative values?	Financial accuracy

ID	Data Quality Question	Why It Matters
DQ-008	Are item prices and freight values within valid ranges?	Transactional accuracy
DQ-009	Are review scores restricted to the expected 1–5 range?	Domain validity
DQ-010	Are there duplicate order-item keys (order_id, order_item_id)?	Transaction integrity
DQ-011	Are there duplicate review records or inconsistent review identifiers?	Review-level integrity
DQ-012	Are order timestamps logically ordered?	Temporal integrity
DQ-013	Are missing delivery timestamps consistent with order status?	Business-rule validation
DQ-014	Are product weights and dimensions valid, including zero and missing values?	Product and logistics analysis
DQ-015	Are product categories missing or inconsistent with the category translation table?	Category-level analytical accuracy
DQ-016	Are geolocation records appropriately structured for ZIP-prefix mapping?	Prevents incorrect geographic joins
DQ-017	Do geographic mappings contain conflicting city/state information for the same ZIP prefix?	Geographic integrity
DQ-018	Are there orphan records across any of the major entity relationships?	Prevents loss or duplication during joins

15. Profiling Conclusion

The initial profiling establishes that the Olist dataset contains multiple interconnected entities operating at different levels of granularity, including customers, orders, order items, payments, products, sellers, reviews, and geolocation records. The dataset provides sufficient transactional, product, customer, and geographic information to support customer segmentation, retention analysis, customer value assessment, and operational performance analysis.

The profiling also identified several structural characteristics requiring validation before analytical modeling. These include the distinction between customer_id and customer_unique_id, multiple records per order in the order-item and payment tables, missing operational timestamps, incomplete product attributes, and repeated geographic observations at the ZIP-prefix level.

These findings will guide the subsequent Data Quality Assessment, where the identified risks will be systematically tested. The results will then be translated into explicit Silver-layer cleaning and transformation rules, ensuring that downstream analytical datasets are reliable, consistent, and aligned with the intended business grain.